When Competition Compels Change: Trade, Management, and Productivity

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October 1, 2026

REStud Tour
Northwestern University

Gains from Trade “Inside the Firm”

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- Selection: competition drives out laggard firms (Melitz, 2003)
- Within-firm "Slack" or "X-inefficiency"
  - *"Absent competitive pressure, firms may not use their resources efficiently"* (Leibenstein, 1966)
- **This paper:** competition compels firms to improve management quality and firm organization
  —⟶ accompanied by within-firm productivity growth
- In today's global context, understanding how trade shapes organizations is increasingly urgent

Management Quality and Firm Organization

Management Quality and Firm Organization



Shri George Jacob Muthoot

Group Chairman
The Muthoot Group of Companies



V.A. George
Independent Director
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George Alexander
Deputy Managing Director
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George Muthoot Jacob
Deputy Managing Director
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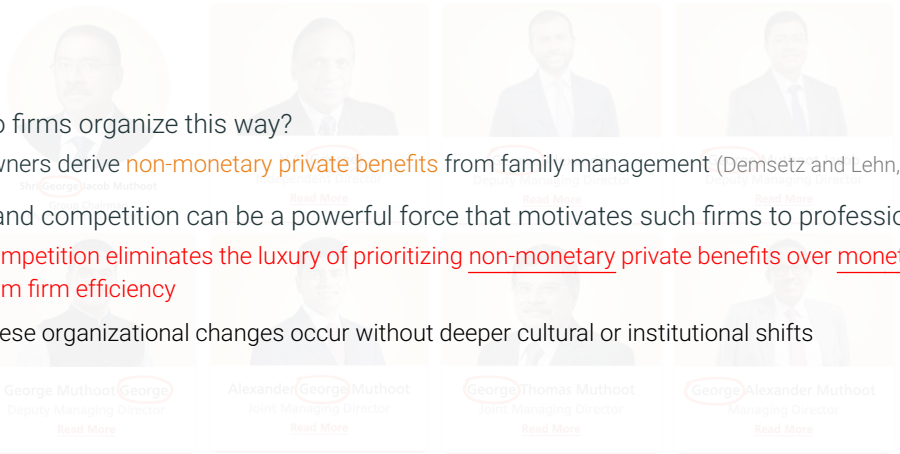


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Joint Managing Director
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Management Quality and Firm Organization: The House of Georges



- Why do firms organize this way?
  - Owners derive non-monetary private benefits from family management (Demsetz and Lehn, 1985)
- Trade and competition can be a powerful force that motivates such firms to professionalize
  - Competition eliminates the luxury of prioritizing non-monetary private benefits over monetary gains from firm efficiency
  - These organizational changes occur without deeper cultural or institutional shifts

Foreign Competition Induces Family Firms to Professionalize

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2. **Natural Experiment:** large, product-specific import competition shock → event-study design

3. **Quantitative Framework:** How much does the within-firm response matter in the aggregate?

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3. **Quantitative Framework:** How much does the within-firm response matter in the aggregate?
   - Competition compels low productivity firms to professionalize to avoid exit
   - Impact on aggregate productivity and welfare is ambiguous in theory:
     a. Professionalizing firms become more productive
     b. Saving laggards weakens selection
     c. Saving laggards preserves domestic variety

Contribution to the Literature

- **Competition and Firm Productivity:** Hicks (1935); Leibenstein (1966); Nalebuff and Stiglitz (1983); Hart (1983); Schmidt (1997); Raith (2003); Nickell (1996); Schmitz (2005); Holmes & Schmitz (2010); Backus (2020)

→ Open the black box of *"X-inefficiency"*: managerial upgrading

- **Managerial Practices:** Bloom and Van Reenen (2007); Bloom et al. (2012, 2013)

→ Why do poor managerial practices persist despite high returns? Can competition lead to better management?

- **Pro-competitive Gains From Trade:**
  Across-firm resource reallocation: Melitz (2003); ACR (2012); Edmond et al. (2014); Arkolakis et al. (2019)
  Within-firm productivity gains: Caliendo and Rossi-Hansberg (2012); Caliendo et al. (2015, 2020); Bustos (2011); Akcigit Melitz (2022); Bloom et al. (2016); Topalova Khandelwal (2011); Amiti and Konings (2007); Bernard et al. (2011); Dhingra (2013) Atkin Khandelwal (2017); Alfaro-Urena et al. (2022) Autor et al. (2020); Chen Steinwender (2021)

→ Organizational channel (talent of managers) of within-firm productivity gains from trade

- **Family Firms:** Bertrand Schoar (2006); Burkart et al. (2003); Bennedsen et al. (2007); Pérez-González (2006); Villalonga Amit (2007), Bertrand et al. (2008); Caselli Gennaioli (2013); Bandiera et al. (2018); Lemos Scur (2018); Akcigit et al. (2021)

→ Response of family firms to globalization

Outline

1. Introduction
2. Natural Experiment
3. Data
4. Empirical Results
5. Model
6. Calibration & Counterfactuals

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Removal of Quantitative Restrictions (QRs): Large, Product-Specific Trade Shock

1950s- India used **Quantitative Restrictions (QRs)** to protect domestic firms

1990s: First-generation reforms in 1991 removed some QRs; many remained

- 3000 HS 8-digit products, almost 30% of all tariff lines
- Mostly final consumer goods
- QRs continued by exploiting GATT Article XVIII:B (QRs on weak BoP grounds)

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1997: Second-generation WTO-imposed reforms: removal of all remaining QRs

- After complaints to WTO by the US and EU

2000s: Imports of liberalized products soared after the reform. No impact on exports.

Indian Response | HS Product Code Example | Media Coverage | Empirical Strategy | Timeline and Scope | Impact on Trade

Unusually Narrow Scope of India's Second-Generation Trade Reforms

- Variation in import competition across the product space and time
- Externally imposed: hinged on IMF technical assessment of India's BoP
- Unlike first-gen reforms, no concurrent domestic deregulation → isolate trade reform impact
- No change in export policy → isolate import competition impact
- Affected mostly consumer goods: limited impact on firms' input costs

HS Product Code Example

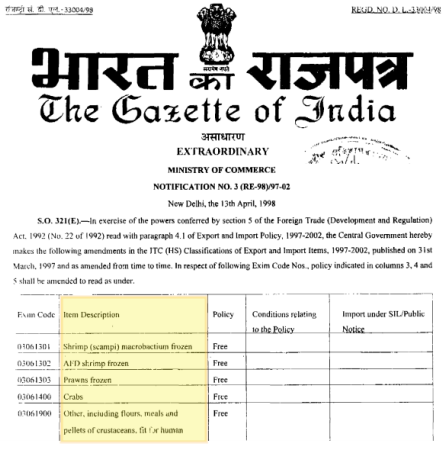
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Data Construction

1. QR removal: Identify affected products
   - More precise treatment assignment
   - ∼ 200 industries in past work vs. 10k+ products

Digitized > 1,000 pages of Customs Notifications



रजिस्ट्री सं. डी. एल.-33004/98

REGD. NO. D. L.-33004/98

भारत का राजपत्र

The Gazette of India

असाधारण

EXTRAORDINARY

MINISTRY OF COMMERCE

NOTIFICATION NO. 3 (RE-98)/97-02

New Delhi, the 13th April, 1998

S.O. 321(E).—In exercise of the powers conferred by section 5 of the Foreign Trade (Development and Regulation) Act, 1992 (No. 22 of 1992) read with paragraph 4.1 of Export and Import Policy, 1997-2002, the Central Government hereby makes the following amendments in the ITC (HS) Classifications of Export and Import Items, 1997-2002, published on 31st March, 1997 and as amended from time to time. In respect of following Exim Code Nos., policy indicated in columns 3, 4 and 5 shall be amended to read as under:

Exim Code	Item Description	Policy	Conditions relating to the Policy	Import under SIL/Public Notice
03061301	Shrimp (scampi) macrobactium frozen	Free		
03061302	AFD shrimp frozen	Free		
03061303	Prawns frozen	Free		
03061400	Crabs	Free		
03061900	Other, including flours, meals and pellets of crustaceans, fit for human	Free		

Source: Archives, Ministry of Commerce, Government of India

Data Construction

1. QR removal: Identify affected products
   - More precise treatment assignment
   - $\sim$ 200 industries in past work vs. 10k+ products
2. QR removal: Identify affected firms

Novel Product Concordance

Firm Data		Customs Notifications (HS-8)	
3008040800	Sunflower seed oil	15121910	Sunflower oil edible grade
3008040804	Sunflower seed oil, refined		
4012080400	Suitcases	42021204	Plastic moulded suit-cases
5024200404	Distempers	32100001	Distempers
6308361216	Washing Machines/Laundry Mach	84501100	Fully - automatic washing machines (upto 10kg)
		84501200	Other washing machines with built-in centrifugal drier (upto 10kg)
		84501300	Other washing machines (up to 10kg)

6,000 products reported by Prowess firms

$\longleftrightarrow$

10,000 8-digit HS products in customs notifications

Data Construction

1. QR removal: Identify affected products
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2. QR removal: Identify affected firms
3. Impact (financial): CMIE Prowess:
   - Product scope
   - Balance sheet + financial statements Summary Stats

Data Construction

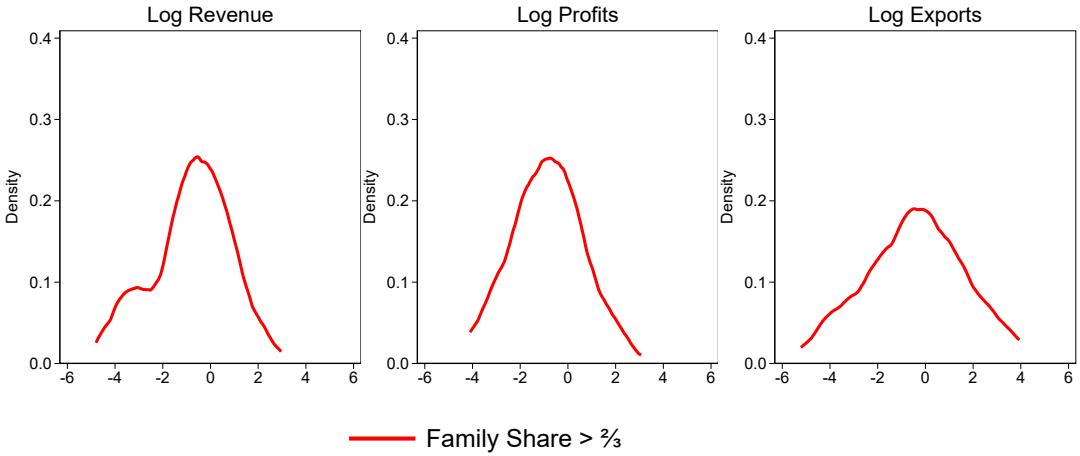
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4. Impact (managerial): MCA Admin Data
   - Family ties among top execs/directors
   - Tenure history from 1980s
   - $>$ 6 million directors

Family Ties Among Top Executives

Name	Father's Name	Executive Director	Sex	Age	Place of Birth
A Khosla	D K Khosla	Yes	M	44	Delhi
M Khosla	D K Khosla	Yes	M	40	Delhi
P Khosla	D K Khosla	Yes	M	39	Delhi
N Khosla	N K Khosla	Yes	M	38	Delhi
D K Khosla	K L Khosla	Yes	M	72	Delhi
N K Khosla	K L Khosla	Yes	M	69	Jhelum
M P Gupta	P D Gupta	No	M	70	Delhi
V K Sood	H R Sood	No	M	67	Lahore
M L Mangla	T Chand	No	M	70	Sangrur

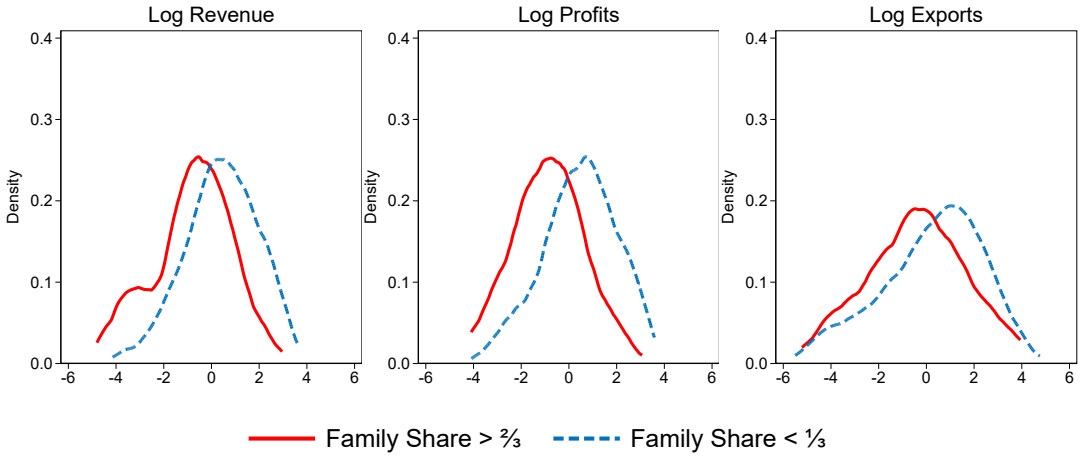
Note: Names abbreviated and anonymized. Based on a real firm.
Source: Confidential administrative records, Corporate Affairs Ministry

Baseline Descriptive Statistics: Family Firms are Smaller and Less Profitable



Model: Baseline Equilibrium | Productivity Distribution | Estimation

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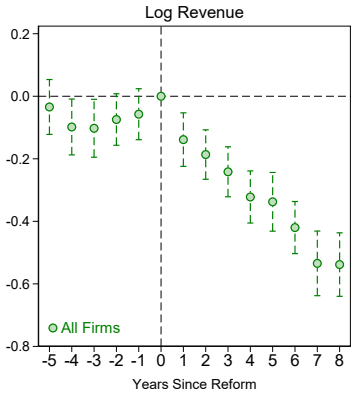
Event-study Design

$$Y_{it} = \sum_{k=\underline{T}}^{\overline{T}} \theta_k D_{it}^k + \delta_i + \lambda_{jt} + \varepsilon_{it}$$

- $D_{it}^k := \mathbb{1}[t = s_i + k]$ are event-time dummies
  - s_i: first year in which QRs are removed on firm i's highest-revenue (8-digit) product
  - Robust to alternate treatment assignment rules
- δ_i: firm FE
- λ_{jt}: year $\times$ 3-digit industry FE

- Sun and Abraham (2021) estimator. Only using never-treated firms as controls

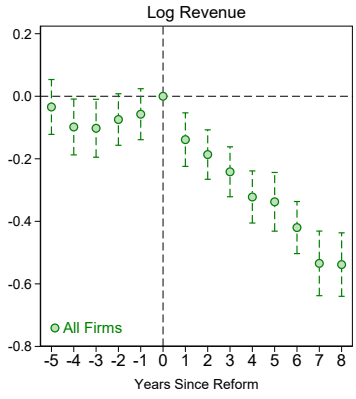
❶ First Stage: Firms Contract

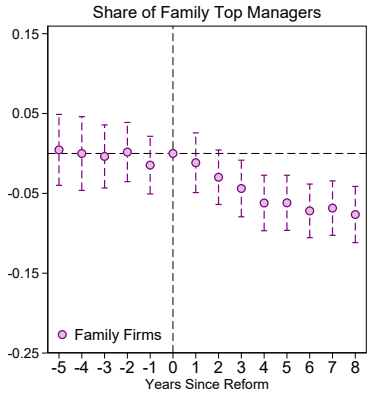


Treated vs. Control Group | Wage Bill and Inputs | Alt Specs | Fam/Prof Firms Separately | Bounds Financial | Bounds Management | Survivors

Number of Managers | Family Majority | Place and State of Birth | Directors' Age Distribution | Ownership | Exit | Model: Baseline Equilibrium | Model: Comparative Statics

❶ First Stage: Firms Contract ❷ Laggards Professionalize

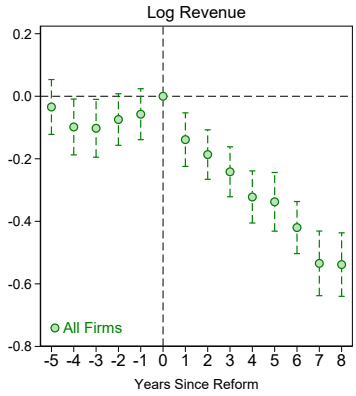


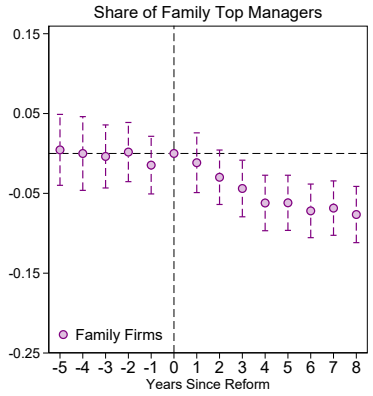


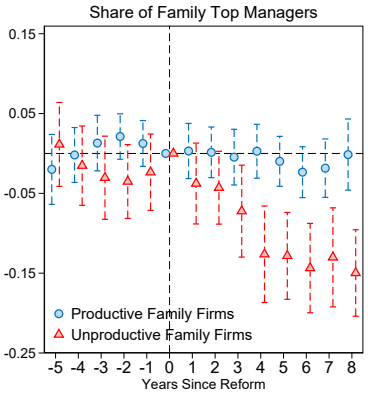
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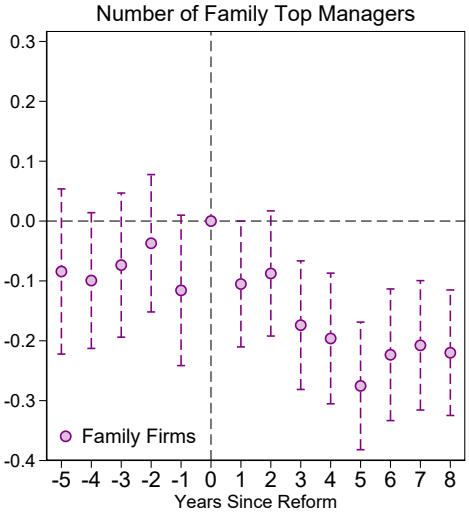


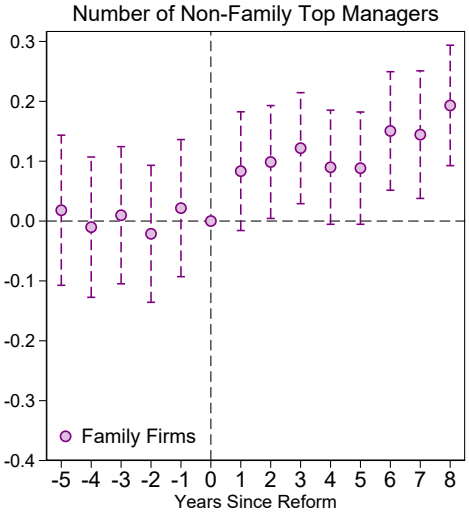


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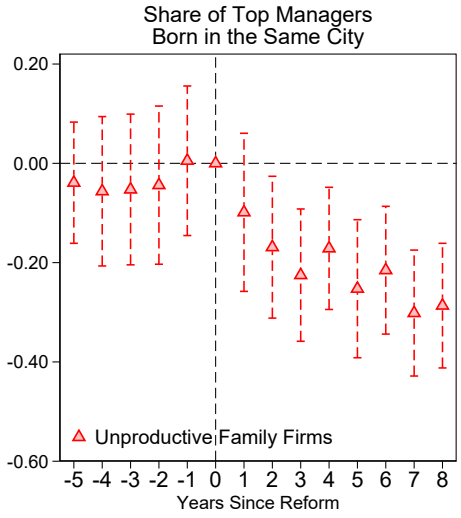
❷ Laggards Professionalize: Number of Family and Professional Executives

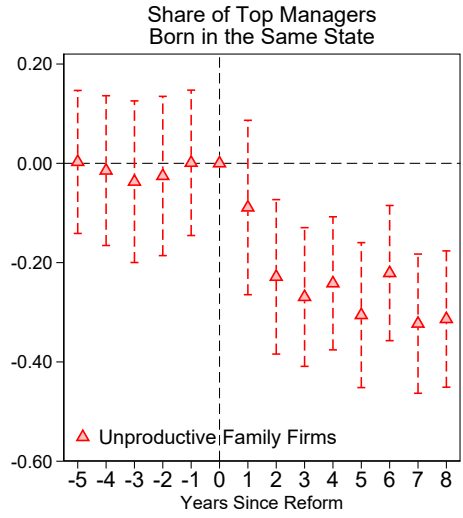




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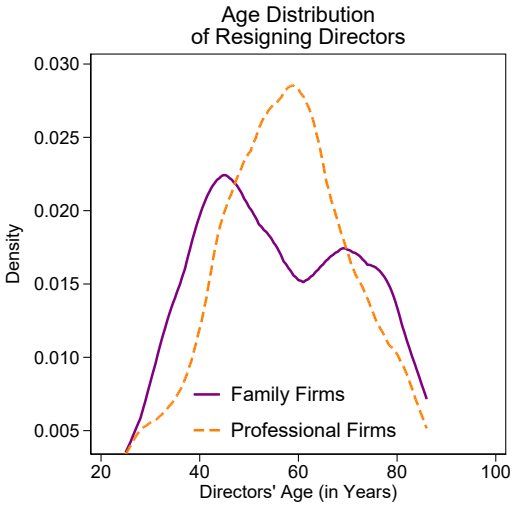
❷ Laggards Professionalize: City and State of Birth

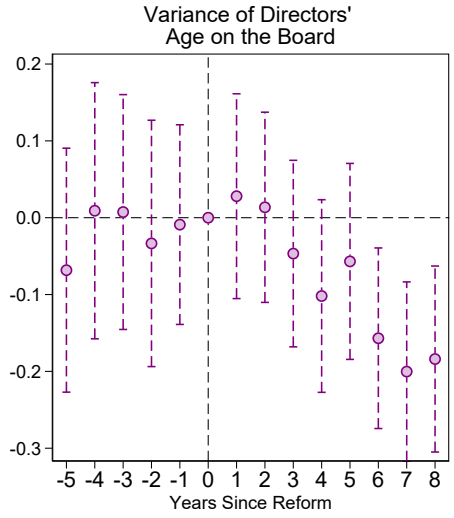




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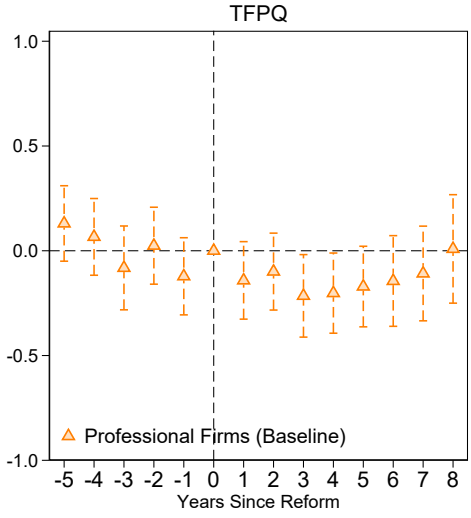
❷ Laggards Professionalize: Compression of the Executives' Age Distribution



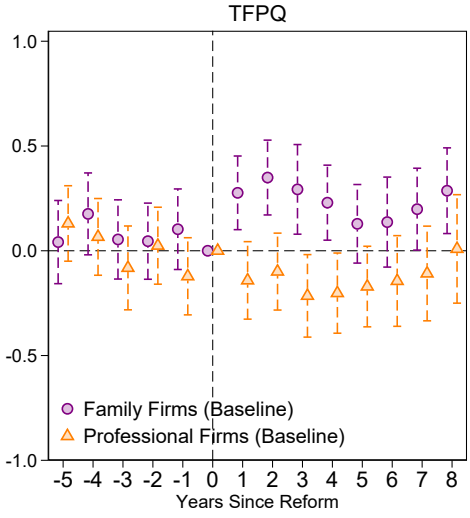


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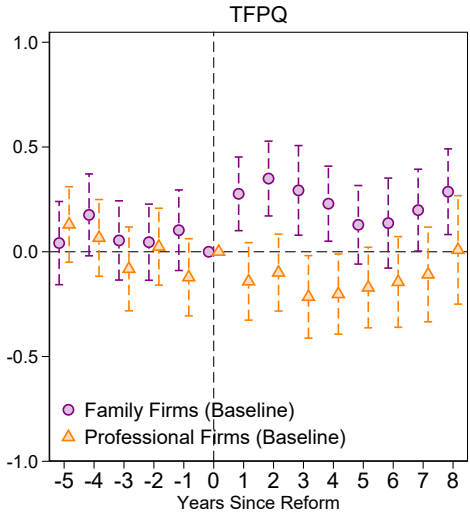
❸ Firms that Professionalize Report Higher Quantity Productivity

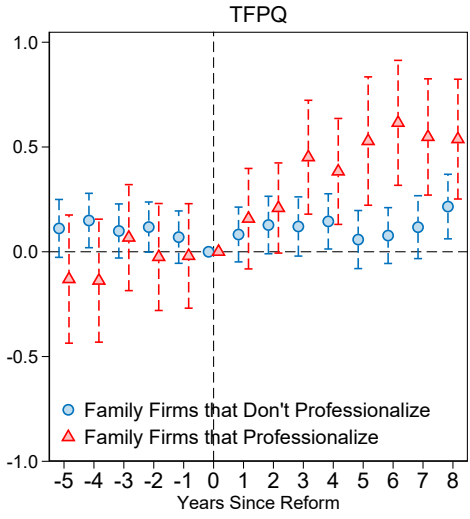


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Note: A family firm is defined as having professionalized if its share of family top managers was lower in $t = 8$ as compared to $t = 0$.

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Management Choice

- Monopolistic competition with CES demand, heterogeneous firm productivity, z

Preferences | Technology

$$\text{Entrants draw } z \sim G(z) \longrightarrow \text{Firm's payoff} = \begin{cases} \pi(z) + \mathcal{B} & \text{if firm stays family} \\ \pi(\gamma z) & \text{if firm professionalizes} \\ 0 & \text{if firm exits} \end{cases}$$

- $\mathcal{B}$: owners derive non-monetary private benefits from running a firm as a family firm
  (Demsetz and Lehn, 1985; Burkart, Panunzi, and Shleifer, 2003; Bertrand and Schoar, 2006; Hurst and Pugsley, 2015)
- $\gamma > 1$: gains from professionalizing firm management. Absorbing state — Reputation Cost
- Firm owners are hand-to-mouth: $\implies$ exit if $\pi(z) < 0$

Equilibrium

Management Choice

- Monopolistic competition with CES demand, heterogeneous firm productivity, z

Preferences | Technology

$$\text{Entrants draw } z \sim G(z) \longrightarrow \text{Firm's payoff} = \begin{cases} \pi(z) + \mathcal{B} & \text{if firm stays family} \\ (1-\tau) \cdot \pi(\gamma z) - f_d & \text{if firm professionalizes} \\ 0 & \text{if firm exits} \end{cases}$$

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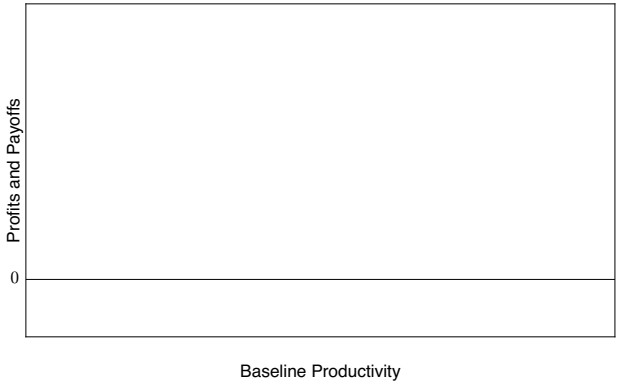
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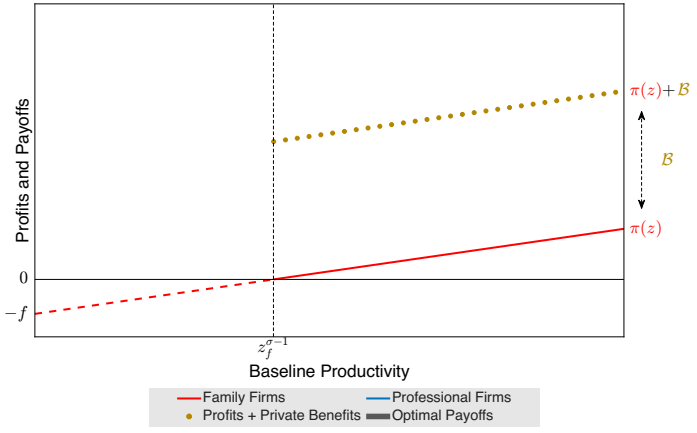
Equilibrium

Firm Profits and Management Choice:



Note: Baseline productivity: before decision to professionalize.

Firm Profits and Management Choice: Family Firms



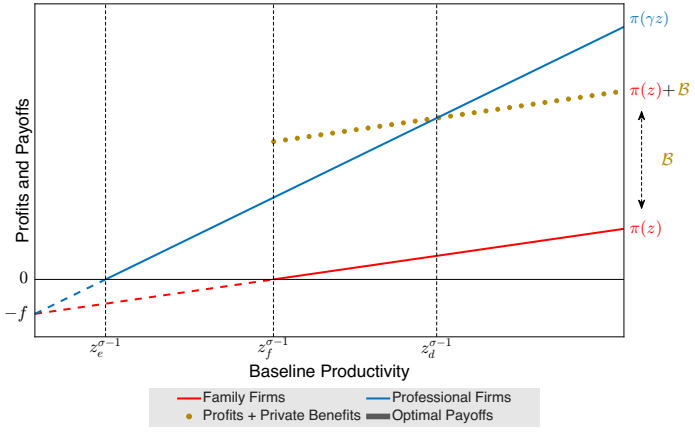
Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

Note: Baseline productivity: before decision to professionalize.

Exit | Contracting Frictions | Productivity Distribution | Data | Event Studies | Estimation

Firm Profits and Management Choice: Professional Firms



Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

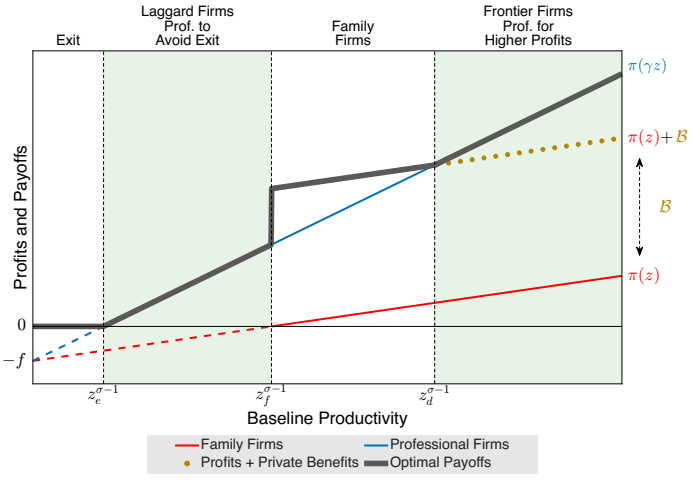
$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before decision to professionalize.

Exit | Contracting Frictions | Productivity Distribution | Data | Event Studies | Estimation

Firm Profits and Management Choice: Optimal Payoff



Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

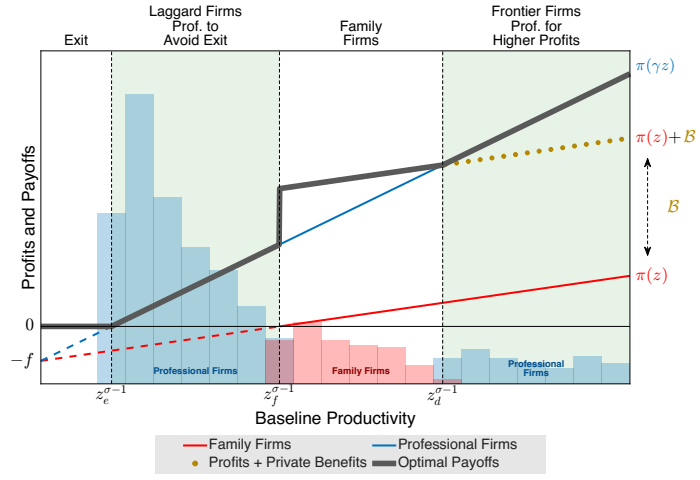
$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before decision to professionalize.

Exit | Contracting Frictions | Productivity Distribution | Data | Event Studies | Estimation

Firm Profits and Management Choice: Baseline Productivity Distribution



Productivity Thresholds

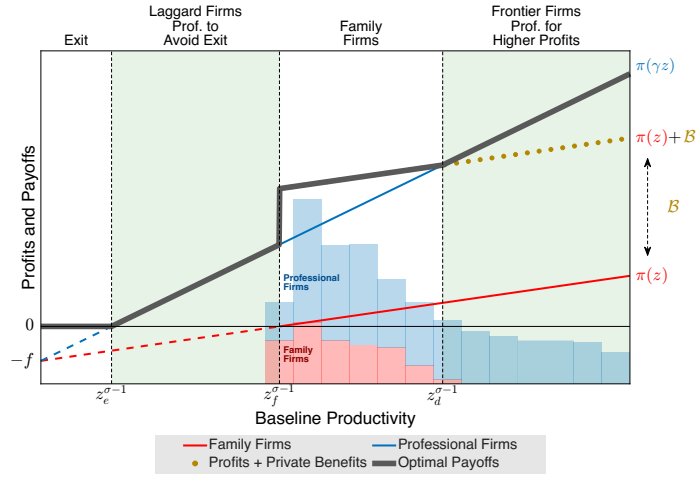
$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before the decision to professionalize. Model: Baseline Equilibrium Data

Firm Profits and Management Choice: Observed Productivity Distribution



Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

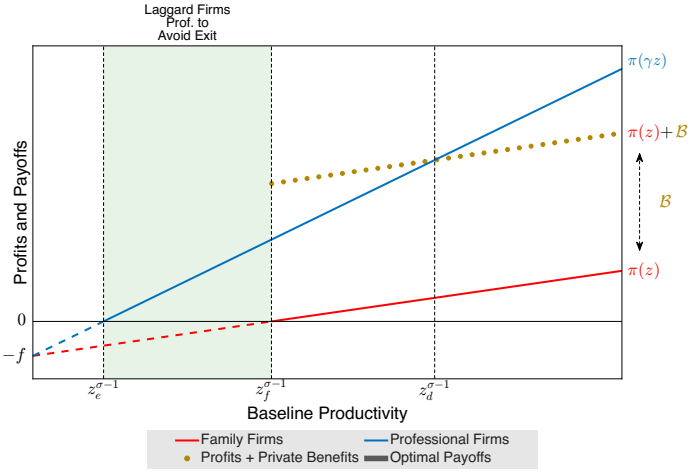
$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before the decision to professionalize.

Model: Baseline Equilibrium | Data

Family Firms: Response to ↑ Import Competition, ↓ Market Demand



Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

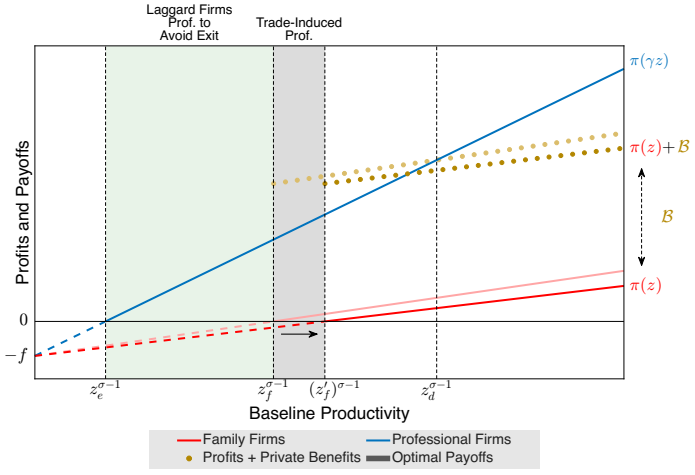
$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before the decision to professionalize.

Import Competition | Contracting Frictions | Reputation Cost | Event Studies

Family Firms: Response to ↑ Import Competition, ↓ Market Demand



Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

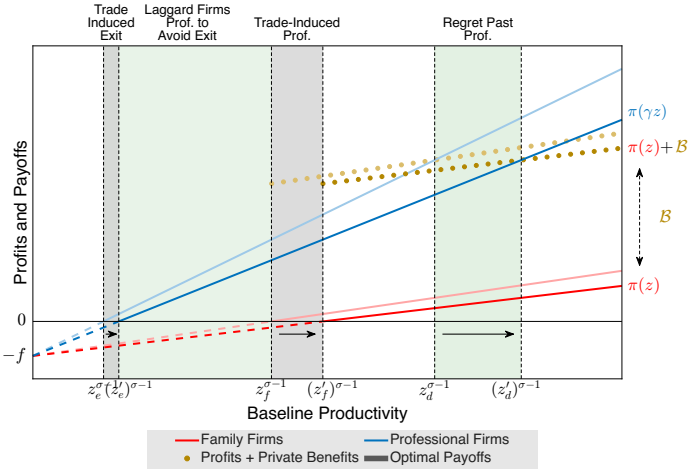
$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before the decision to professionalize.

Import Competition | Contracting Frictions | Reputation Cost | Event Studies

Professional Firms: Response to ↑ Import Competition, ↓ Market Demand



Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before the decision to professionalize.

Import Competition | Contracting Frictions | Reputation Cost | Event Studies

Outline

1. Introduction
2. Natural Experiment
3. Data
4. Empirical Results
5. Model
6. **Calibration & Counterfactuals**

Calibration

$$\Theta = \{\sigma, k, \gamma, \mathcal{B}\}$$

- $\sigma = 4$
- $k = 3.5 \longrightarrow$ upper-tail slope in a log-rank vs. log-productivity QQ plot Head, Mayer, Thoenig (2014)
- $\mathcal{B}, \gamma$ jointly determine management choice Moment Conditions
- Calibrate the model to India's pre-policy economy

Parameter	Identifying moment	Estimate
$\mathcal{B}$	Share of family firms	
γ	Log mean revenue gap: professional minus family firms	1.23

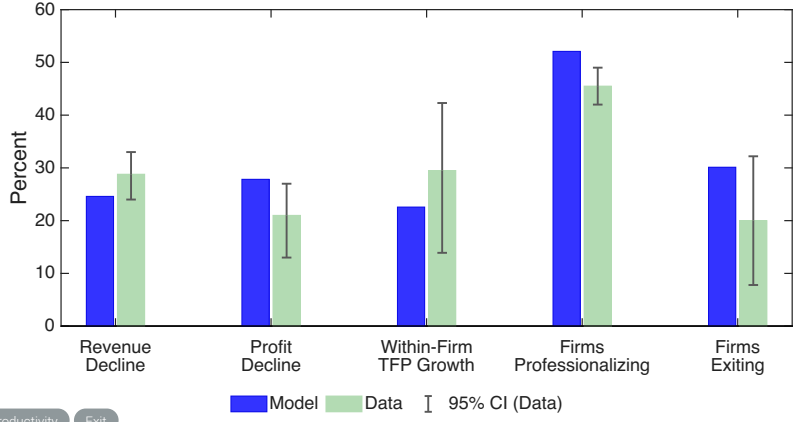
Data Model: Baseline Equilibrium

Comparison of Model and Data Estimates

- **Policy experiment:** increase import penetration from its pre- to post-policy level
- Re-solve the post-shock allocation
- Compare model-implied changes in untargeted moments to event-study results

Comparison of Model and Data Estimates

- **Policy experiment:** increase import penetration from its pre- to post-policy level
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- Compare model-implied changes in untargeted moments to event-study results



Revenue and Profits | Productivity | Exit

Aggregate Productivity and Welfare

	Benchmark model Professionalization permitted	Counterfactual No new professionalization	Difference (1) – (2)
Panel A. Aggregate productivity change			
Total			
Between-firm selection			
Within-firm productivity			
Panel B. Welfare change relative to baseline			
Consumption welfare			
Welfare including private benefits			

Details

Aggregate Productivity and Welfare

	Benchmark model Professionalization permitted	Counterfactual No new professionalization	Difference (1) – (2)
Panel A. Aggregate productivity change			
Total	11.45%		
Between-firm selection	10.62 pp		
Within-firm productivity	0.83 pp		
Panel B. Welfare change relative to baseline			
Consumption welfare			
Welfare including private benefits			

Details

Aggregate Productivity and Welfare

	Benchmark model Professionalization permitted	Counterfactual No new professionalization	Difference (1) − (2)
Panel A. Aggregate productivity change			
Total	11.45%	15.68%	−4.23 pp
Between-firm selection	10.62 pp	15.68 pp	−5.06 pp
Within-firm productivity	0.83 pp	0.00 pp	+0.83 pp
Panel B. Welfare change relative to baseline			
Consumption welfare			
Welfare including private benefits			

Details

Aggregate Productivity and Welfare

	Benchmark model Professionalization permitted	Counterfactual No new professionalization	Difference (1) − (2)
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Panel B. Welfare change relative to baseline			
Consumption welfare	8.48%	7.36%	+1.13 pp
Welfare including private benefits			

Details

Aggregate Productivity and Welfare

	Benchmark model Professionalization permitted	Counterfactual No new professionalization	Difference (1) − (2)
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Within-firm productivity	0.83 pp	0.00 pp	+0.83 pp
Panel B. Welfare change relative to baseline			
Consumption welfare	8.48%	7.36%	+1.13 pp
Welfare including private benefits	7.76%	6.71%	+1.04 pp

Details

Conclusion

- Trade drives aggregate growth by improving laggard firms' productivity
- Empirical lens here is family firms, the mechanism is more general
- Protection from competition can sustain taste-based deviations from merit (Becker 1957)
  - along multiple dimensions: kinship, caste, religion, race, or gender
- Competition may help restore meritocracy
- These organizational changes occur without deeper institutional shifts

Thank you!

Ananya Kotia

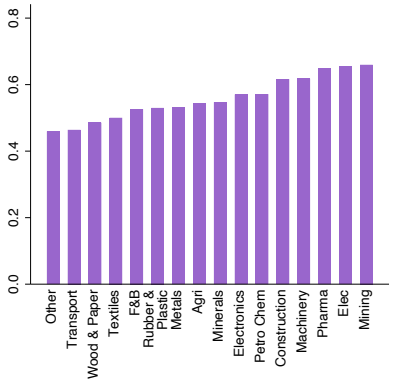
Stanford University

ananyakotia.com

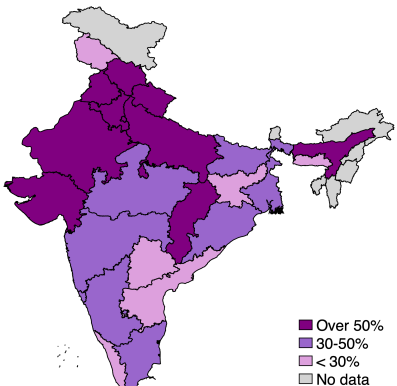
Family Firms are Widespread in India

Family Firms: at least two family members on board

Share of Family Firms across Industries



Share of Family Firms across States

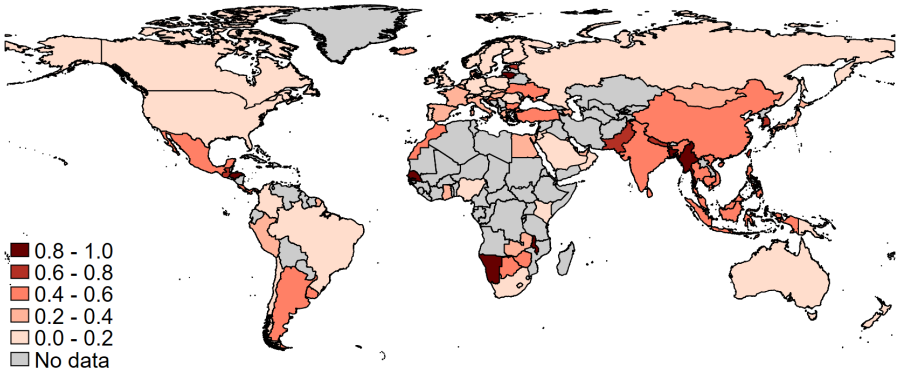


Source: Ministry of Corporate Affairs and CMIE Prowess$_{dx}$

Back

Family Firms are Widespread in Developing Countries

Share of Directors Sharing a Surname



Source: BoardEx, Wharton Research Data Service (WRDS)

Back

Indian Response to WTO's QR Ruling

"The commerce secretary had written to me to say that they intended to make a case for continuing with QRs on consumer goods for BoP reasons, and he wanted confirmation of support from the Ministry of Finance. I promptly wrote back to say that ***it would be against our national interest because it painted a picture of helplessness that was completely contrary to what we in the Ministry of Finance were projecting to foreign investors****"*

—Montek Singh Ahluwalia, Former Indian Finance Secretary and Deputy Chairman of the Planning Commission

Policy response by India:

- Tariffs maintained : limited room to raise beyond WTO bindings
- Canalization of Grains
- "War Room" & sensitive list: DGFT monitored ~300 consumer goods
- Some anti-dumping (esp. chemicals, steel, consumer goods)

Back

HS Codes Example: Electrical machinery

- **8517:** Telephone sets and other apparatus for voice transmission
  - **861711:** Line telephone sets
    - **85171110:** Push button type
    - **85171120:** Rotary dial type
    - **85171130:** Cordless Telephones
  - **861712:** Cellular network sets
    - **85171210:** Push button type
- **8525:** Transmission apparatus and cameras
  - **852550:** Transmission apparatus
    - **85255010:** Radio broadcast transmitter
    - **85255020:** TV broadcast transmitter
    - **85255040:** Jamming equipment
  - **852580:** Cameras
    - **85258010:** Television Cameras
    - **85258020:** Digital cameras
    - **85258030:** Video cameras recorders

Removal of quantitative restrictions on all imports to transform Indian market

Removal of quantitative restrictions on all imports will transform the Indian market like never before.



PROFUSION OF CHOICE: Shelves like these in a Chennai super store are getting stacked with an array of foreign products

Amma Naana Super Store, TTK Road, Chennai: Heinz spaghetti with sausages in tomato sauce, 220 g for Rs 155; H-Vollmilch low-fat milk with one-year shelf life,1 litre for Rs 66; Sugar-free chocolate biscuits, 184 g for Rs 98...

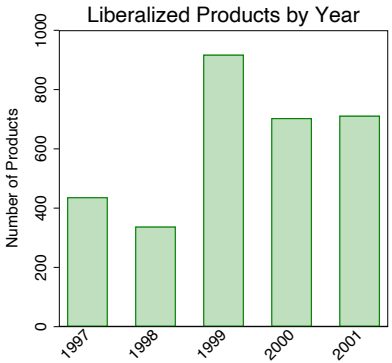
Rustom Stores, Colaba, Mumbai: Blue Bunnyice cream, 1.5 litre for Rs 425; St Martin's iced tea, 240 ml for Rs 25; Langnese honey, 500 g for Rs 165

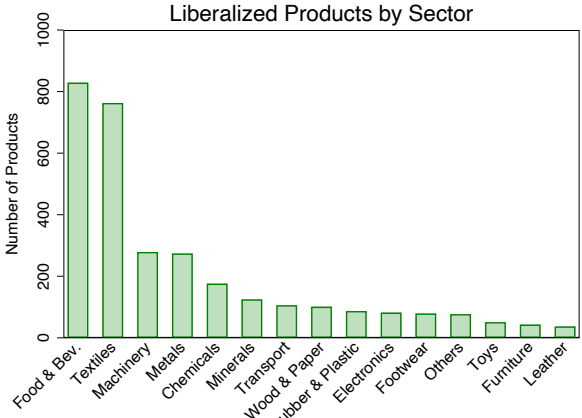
Steak House, Jor Bagh, Delhi: Lake land mayonnaise, 450 g for Rs 98; Laughing Cow cheese, 180 g for Rs 45...

A sample of goodies on the shelves of stores in the three metros on April 3, two days after India abolished all quantitative restrictions (QRs) on imports. On payment of customs duty, anybody can now import any amount of any product. A freedom of economic choice that comes a full 54 years after the freedom of political choice the country attained in 1947.

Source: India Today Magazine, April 16, 2001. Back

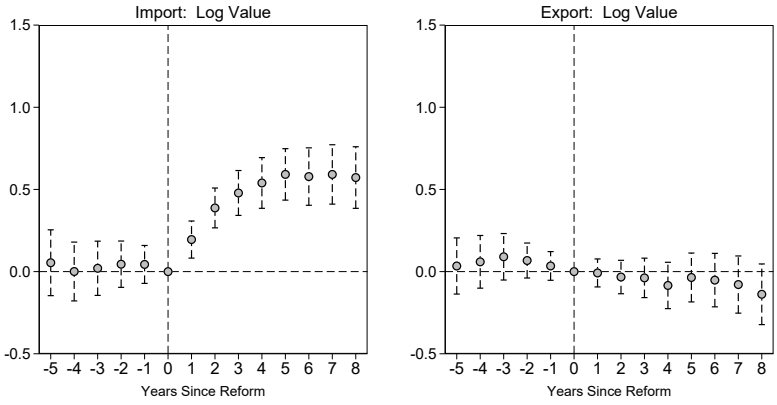
QR-Removal: Timeline and Sectoral Scope





Back

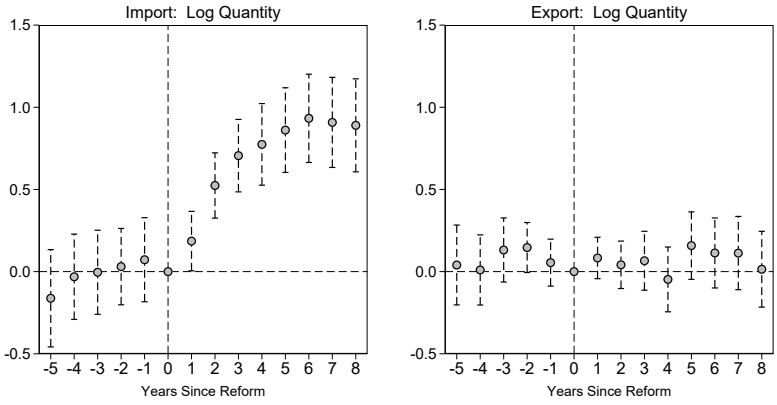
QR-Removal: Imports Soar, No Effect on Exports



$$y_{pt} = \sum_{k=\underline{T}}^{\bar{T}} \beta_k D_{pt}^k + \delta_p + \lambda_{qt} + \varepsilon_{pt}$$

p: 6-digit HS product, q: 4-digit HS product, D_{pt}^k: event-time dummies Quantities Back

QR-Removal: Imports Soar, No Effect on Exports



$$y_{pt} = \sum_{k=\underline{T}}^{\bar{T}} \beta_k D_{pt}^k + \delta_p + \lambda_{qt} + \varepsilon_{pt}$$

p: 6-digit HS product, q: 4-digit HS product, D_{pt}^k: event-time dummies Timeline Values

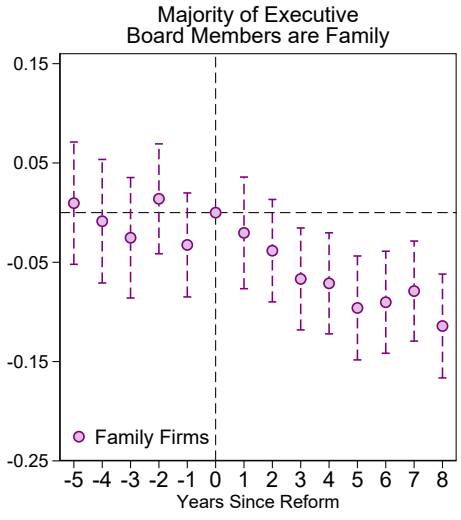
Summary Statistics

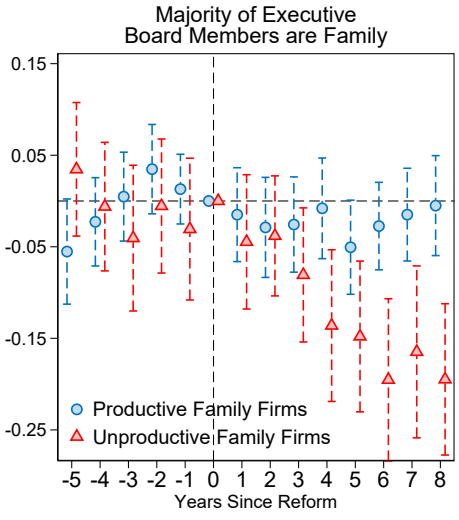
	Obs	p10	Mean	p50	SD
Treated Firms (%)	4,996	0	0	1	0
Firm Age (Years)	83,726	7	26	56	20
Wages	77,724	1	220	326	1,382
Total Assets	82,465	46	4,083	4,983	31,652
Revenues	76,552	19	3,674	4,785	36,687
Expenses on Raw Materials	67,944	13	1,687	2,227	17,783
At least 2 Family Members on Board	4,852	0.00	0.45	1.00	0.50
Family Share on Board	39,644	0.10	0.38	0.75	0.25
Family Share on Executive Board	39,644	0.00	0.64	1.00	0.42

Note: All nominal variables are reported in INR million and are deflated to 2005 prices using the GDP deflator.

Back

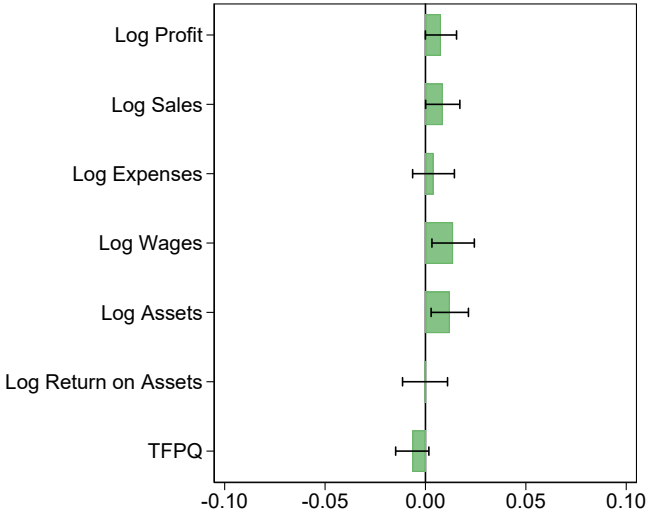
Family Majority on the Executive Board





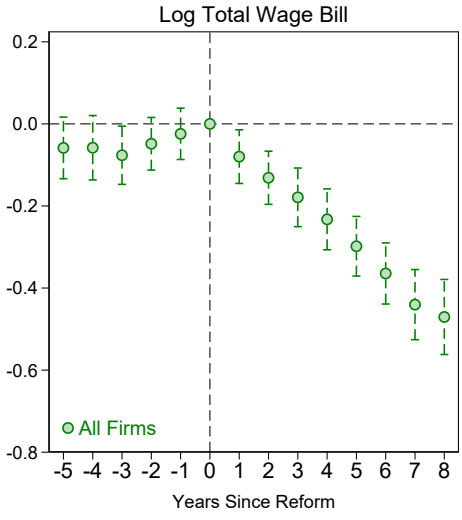
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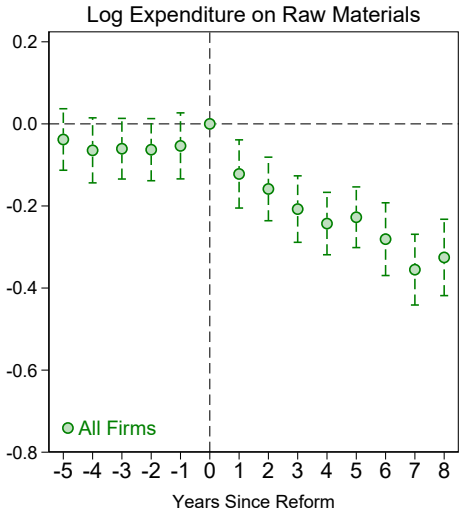
Treated and Control Firms are Similar



Back

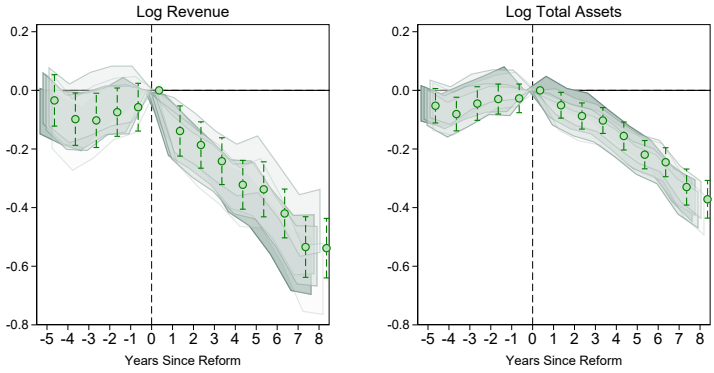
❶ All Firms Contract: Total Wage Bill and Expenses





Back

❶ All Firms Contract: Robustness to Alternative Specifications



Alternative Fixed Effects:

(1) Firm + Year

(2) Firm + (Industry × Year)

(3) Firm + (Industry × Year) + (State × Year)

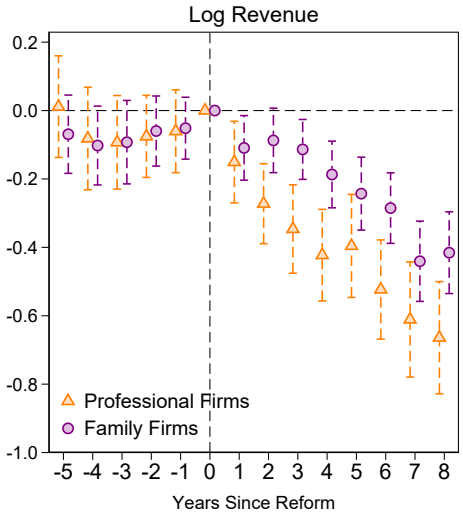
(4) Firm + (Industry × Year) + (District × Year)

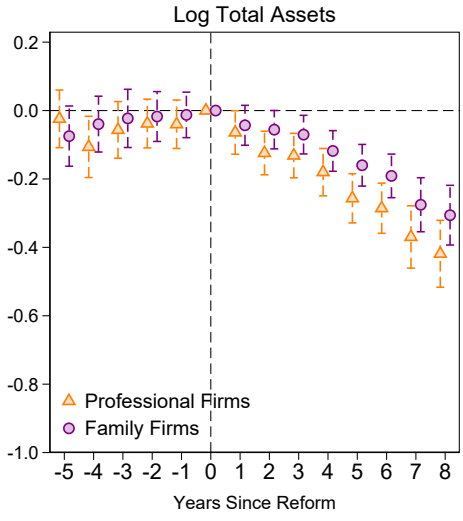
(5) Firm + (Industry × Year × State)

(6) Firm + (Industry × Year × District)

Back

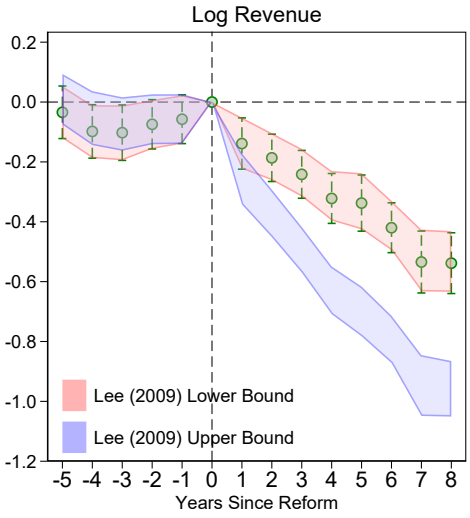
❶ All Firms Contract

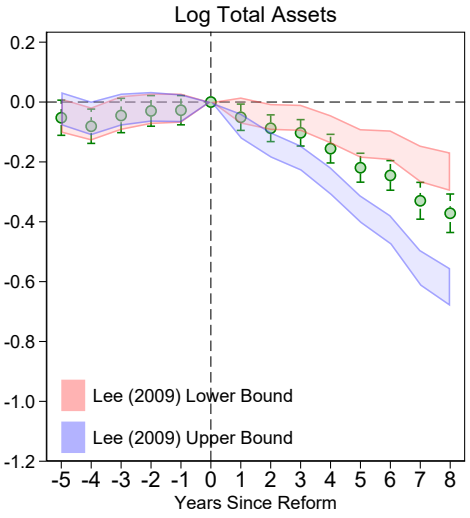




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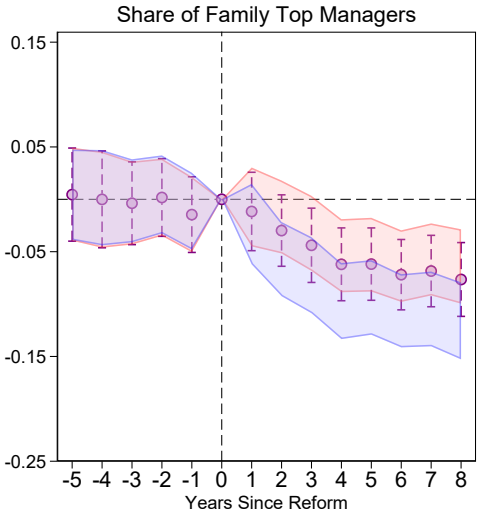
Lee Bounds





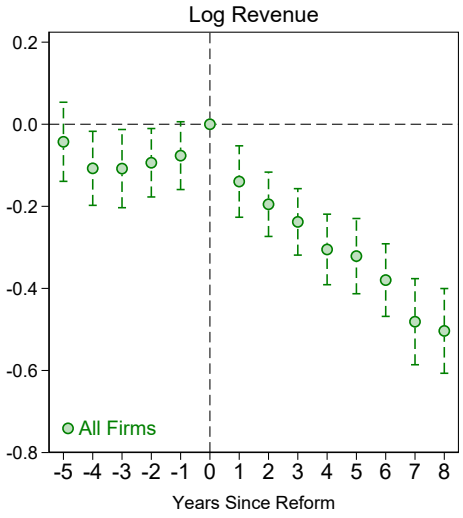
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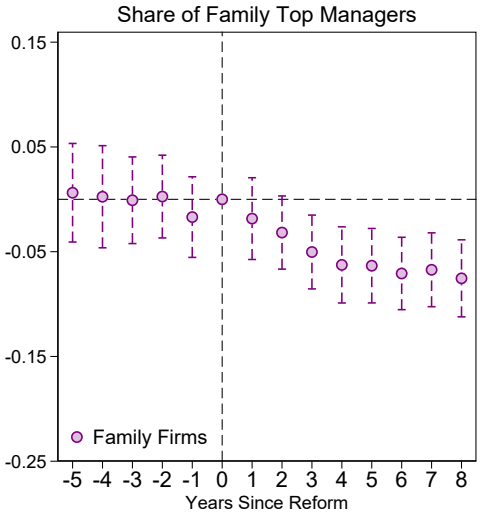
Lee Bounds



Back

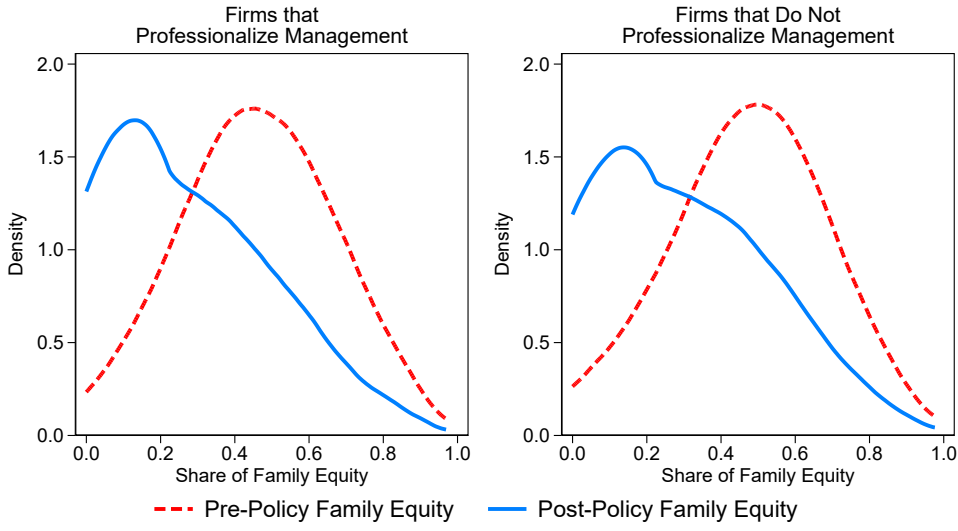
Conditioning on Survivors





Back

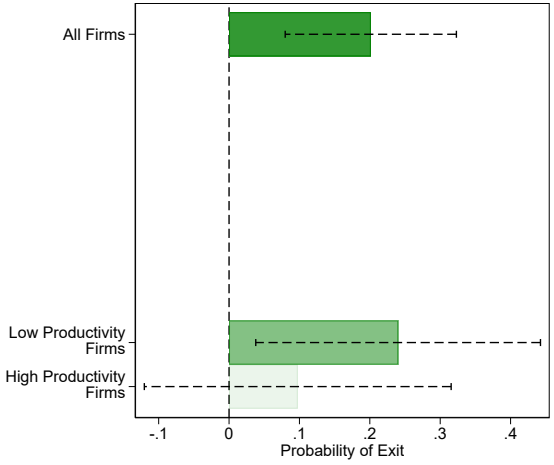
Family Ownership



Back

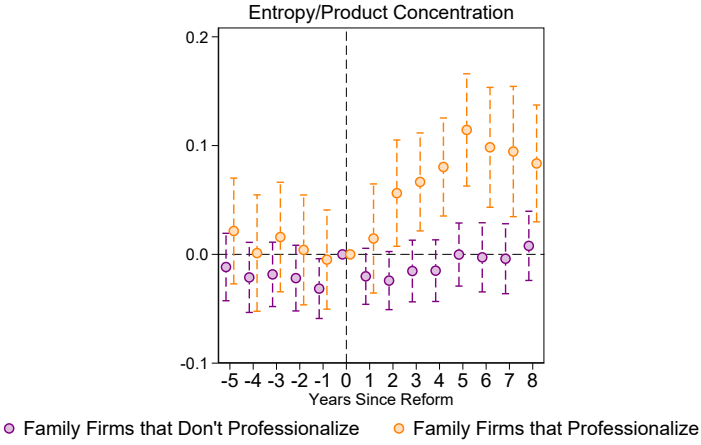
Exit (in the Long Run)

- Cannot confirm exit status or exit year in Prowess data
- Verify exit status by matching firms that drop out of Prowess to MCA admin data
- Long-difference: measure whether firm exits in any year from 2001–2021



Back Model vs Data

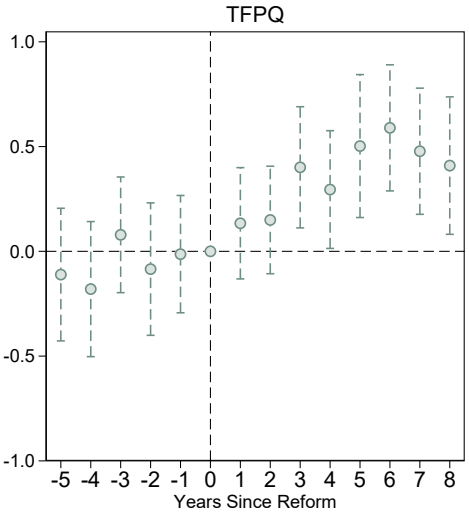
❸ Firms that Professionalize Report Higher Product Concentration



Note: A family firm is defined as having professionalized if its share of family top managers was lower in $t = 8$ as compared to $t = 0$.

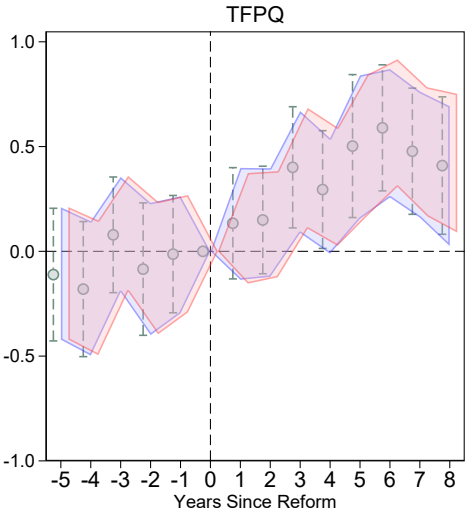
Back

Firms that Professionalize Report Higher Productivity (Triple Difference)



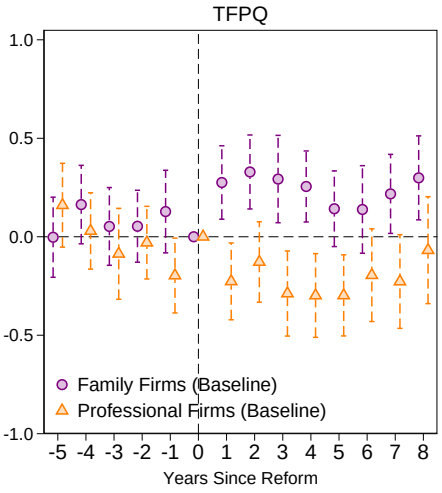
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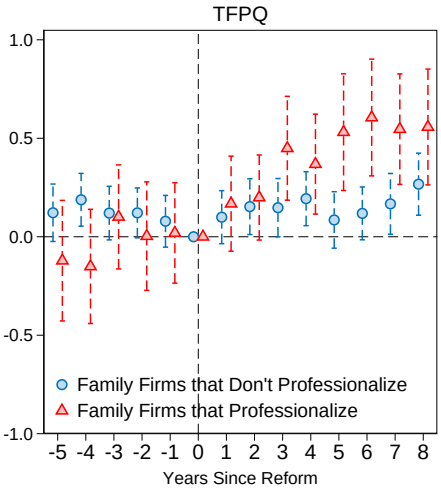
Lee Bounds



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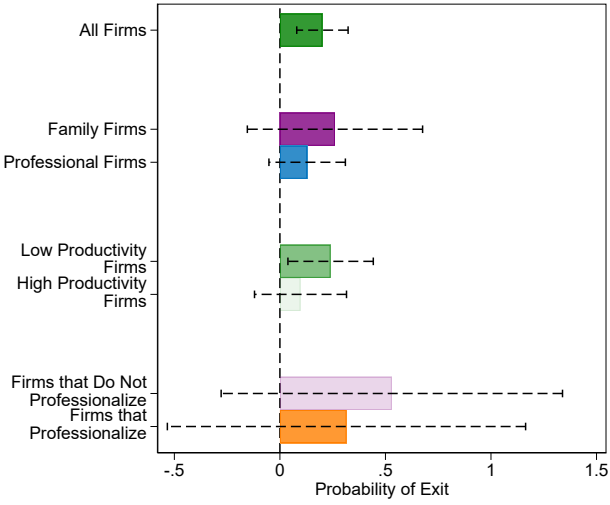
Quantity Productivity: Conditioning on Survivors



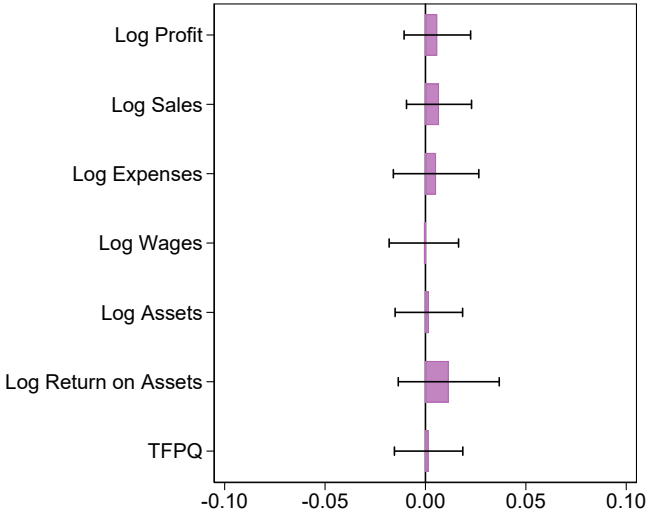


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Exit


All Firms
Family Firms
Professional Firms
Low Productivity Firms
High Productivity Firms
Firms that Do Not Professionalize
Firms that Professionalize
-.5
0
.5
1
1.5
Probability of Exit

Selection Into Professionalizing Management



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Preferences

- Representative consumer

$$\mathcal{U} = X^{1-\alpha} Y^{\alpha} \qquad (0 < \alpha < 1)$$

  - X: freely-traded homogeneous good, with unit labor requirement
  - Y is a CES aggregate of differentiated varieties with elasticity of substitution $\sigma > 1$
- The mass of foreign varieties is M_F
- Consumers spend share α of total expenditure E on good Y
- Demand for each variety y is given by $y = \alpha E P^{\sigma-1} p^{-\sigma}$

Technology

- Firms differ in productivity z and produce differentiated varieties using labor ℓ.
- Constant marginal cost with a fixed overhead cost: $\ell(z) = f + \frac{y}{z}$
- Prices are a markup over marginal cost: $p(z) = \frac{\sigma}{\sigma-1} \cdot \frac{w}{z}$
- Profits: $\pi(z) = Az^{\sigma-1} - wf$ $\left(\text{where } A = \frac{1}{\sigma}\left(\frac{\sigma-1}{\sigma}\right)^{\sigma-1} EP^{\sigma-1}w^{1-\sigma} = \text{market demand}\right)$

Management Choice

Reputational Cost of Switching Back to Family Management



Reuters

World Business Markets Sustainability Legal Breakingviews Technology Investigations More

Infosys CEO resigns after long-running feud with founders

By **Samantha Kareen Nair** and **Sankalp Phartiyal**

August 18, 2017 3:58 PM GMT+1 · Updated 8 years ago

Aa

BENGALURU/MUMBAI (Reuters) - Vishal Sikka, the chief executive brought in to turn around India's Infosys three years ago, resigned on Friday, blaming a "continuous drumbeat of distractions" and a long-running row with the founders over company strategy.

Sikka's resignation spooked investors in India's second-biggest IT services company and sent its shares down nearly 10 percent, wiping $3.45 billion off its market value. The stock touched its lowest level since the start of Sikka's tenure.

Model: Management Choice | Model: Comparative Statics | Profits After Trade

Profits After Trade Liberalization ($t = 1$)

- Switching from professional to family management incurs (reputational) cost κ Example

$$\text{Payoff}\,(z, \mathcal{P}_0, \mathcal{P}_1) = \mathcal{P}_1 \pi(\gamma z) + (1 - \mathcal{P}_1)\left[\pi(z) + B - \kappa \mathcal{P}_0\right]$$

- When $\mathcal{P}_1 = 1$ (the firm upgrades management in $t = 1$), the payoff is $\pi(\gamma z)$ regardless of the past choice
- When $\mathcal{P}_1 = 0$ (the firm is family-managed in $t = 1$), the payoff is:
  - $\pi(z) + B$ if it was family in $t = 0$ (i.e., $\mathcal{P}_0 = 0$)
  - $\pi(z) + B - \kappa$ if it was professionally-managed in $t = 0$ (i.e., $\mathcal{P}_0 = 1$)

Model: Baseline Equilibrium | Model: Comparative Statics | Model: Management Choice

Equilibrium: Free Entry, Exit, and Equilibrium Price Index

- **Free entry:**

$$[1 - G(z_e)] \cdot \frac{1}{\delta}\pi - f_E = 0$$

- **Average profits among surviving firms:**

$$\pi = \frac{\sigma - 1}{k - \sigma + 1} f \cdot \Delta, \qquad \Delta \equiv 1 + \frac{k}{\sigma - 1}\left(\gamma^{\sigma-1} - 1\right)\gamma^{-k}\left(\left(\frac{\mathcal{B}}{f\left(\gamma^{\sigma-1} - 1\right)}\right)^{\frac{\sigma-k-1}{\sigma-1}} - 1\right)$$

- **Exit cutoff:**

$$z_e = \left(\frac{1}{\delta f_E} \cdot \frac{\sigma - 1}{k - \sigma + 1} f\Delta\right)^{1/k}$$

- **Price index (domestic varieties):**

$$P = \frac{1}{\rho\gamma}\left(\sigma f_E\right)^{\frac{1}{\sigma-1}}\left(\frac{\sigma - 1}{k - \sigma + 1} \cdot \frac{f}{\delta f_E} \cdot \Delta\right)^{-1/k}, \qquad \rho \equiv \frac{\sigma - 1}{\sigma}$$

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Equilibrium: Market Clearing and Aggregates

- **Masses of firms.** Let M_E be the mass of entrants and M the mass of active firms:

$$M = \frac{1 - G(z_e)}{\delta} M_E$$

- **Labor demand in differentiated sector.** With revenue R and aggregate operating profits Π:

$$L_Y = \frac{R - \Pi}{w} + M_E f_E$$

  Free entry implies $\Pi = w M_E f_E$

- **Goods-market clearing.** Revenue equals expenditure on differentiated goods:

$$R = E, \qquad E = \alpha I$$

- **Numeraire and income.** Homogeneous good pins down $w = 1$ and total income is the wage bill:

$$I = wL = L \quad \Rightarrow \quad E = \alpha L, \;\; L_Y = \alpha L, \;\; L_X = (1 - \alpha) L$$

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Import Competition: Fall in Aggregate Demand

- Model import competition as an exogenous increase in foreign varieties M_F

$\Longrightarrow$ Lowers the price index

$$P = \left[M_H \int_{z_e}^{\infty} p(z)^{1-\sigma} \frac{g(z)}{[1 - G(z_e)]} dz + M_F \uparrow \int_{z_F}^{\infty} p_F(z)^{1-\sigma} \frac{g(z)}{[1 - G(z_F)]} dz \right]^{1/(1-\sigma)}$$

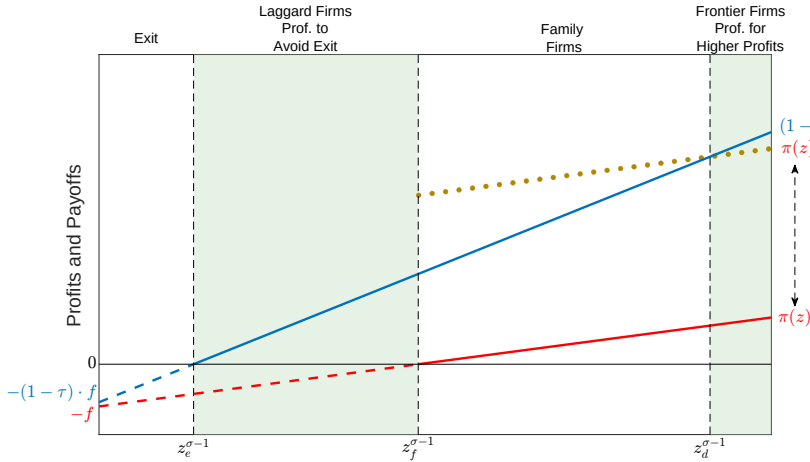
$\Longrightarrow$ Lower aggregate demand for Indian varieties $A = \frac{1}{\sigma} \left(\frac{\sigma - 1}{\sigma} \right)^{\sigma - 1} E P^{\sigma - 1} w^{1-\sigma}$

$\Longrightarrow$ Reduces profits for all domestic firms

$\Longrightarrow$ Thresholds (z_e, z_f, z_d) shift up (more exit, more professionalization)

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Contracting Frictions



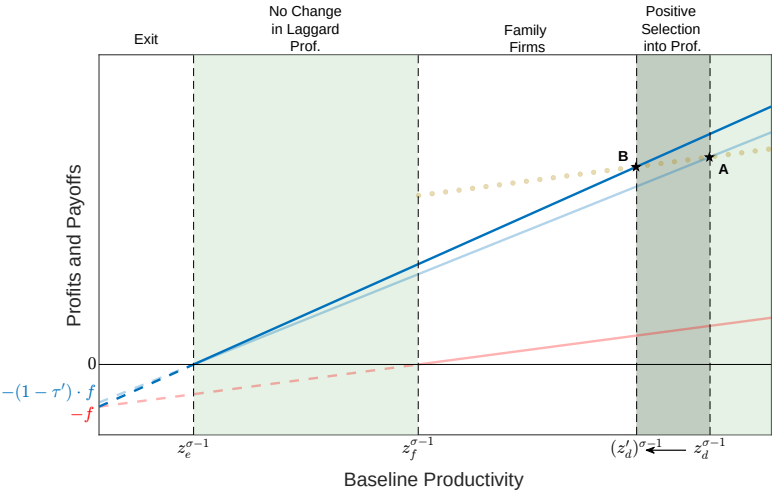
Productivity Thresholds:

- $z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$
- $z_e = \frac{1}{\gamma}\cdot z_f$
- $z_d = \left(\frac{\mathcal{B}-\tau f}{f\left((1-\tau)\gamma^{\sigma-1}-1\right)}\right)^{\frac{1}{\sigma-1}}\cdot z_f$

Note: Baseline productivity: before decision to professionalize.

Model: Baseline Equilibrium | Model: Comparative Statics

Contracting Frictions



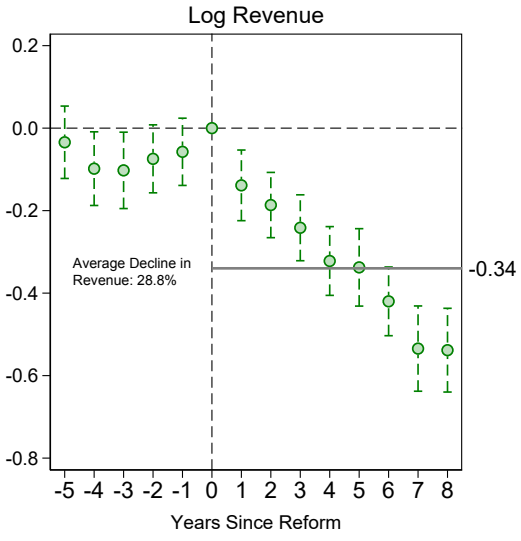
Productivity Thresholds:

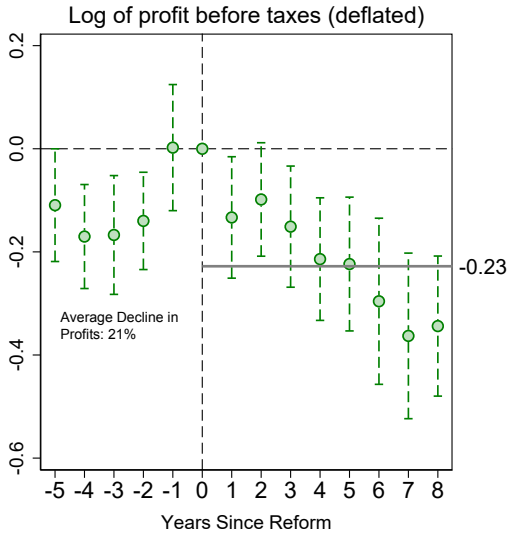
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Model: Baseline Equilibrium | Model: Comparative Statics

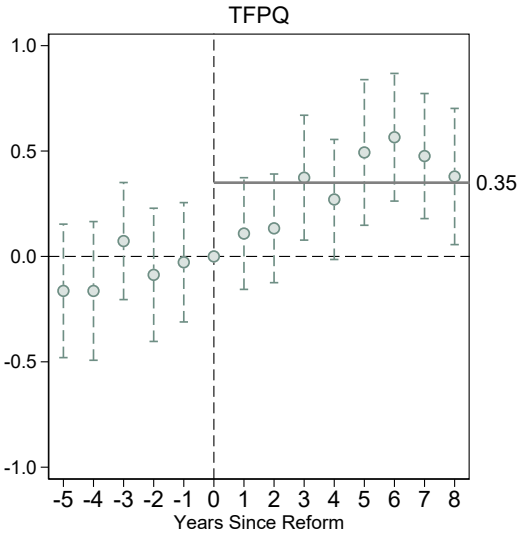
Revenue and Profit Decline





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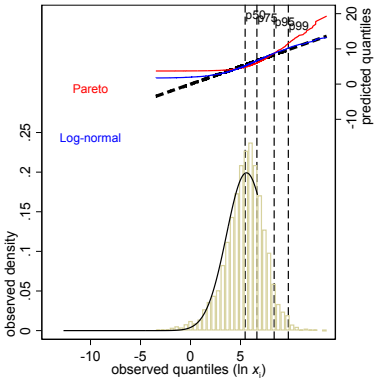
Average Post Treatment Productivity Increases



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Calibrating the Pareto Tail (k)

- **Sample.** Rank firms by revenue, keep 95th–99th pct.
- **Transform.** F_i = empirical CDF $\Rightarrow$ $z_i = -\ln(1 - F_i)$ (linearizes a Pareto tail).
- **QQ-regression.** $\ln x_i = \alpha + \underbrace{\beta}_{=(\sigma-1)/k} z_i + \varepsilon_i.$
- **Result.** $\hat{\beta} = 0.87 \;\Rightarrow\; \hat{k} = \dfrac{\sigma - 1}{\hat{\beta}} = \dfrac{3}{0.87} \approx \mathbf{3.5}.$
- **Robustness.** Cut-offs from 95th to 99th give $k \in [3.4,\ 3.6]$ ✓



QQ-plot of $\ln x_i$ (log revenue) against $z_i = -\ln\left(1 - F_i\right)$ for the top 5% of firms

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Analytical Moment Conditions for $\mathcal{B}, \gamma$

- Log mean revenue gap (professional minus family firms): $(\sigma - 1)\log\gamma + \log\left[\frac{\gamma^{-(\sigma-k-1)} - 1 + \delta^{\sigma-k-1}}{\gamma^{k} - 1 + \delta^{-k}} \cdot \frac{1-\delta^{-k}}{1-\delta^{\sigma-k-1}}\right]$
- Share of family-managed firms: $\gamma^{-k}\left(1 - \delta^{-k}\right)$

where $\delta = \left[\frac{B}{f(\gamma^{\sigma-1}-1)}\right]^{1/(\sigma-1)}$

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Aggregate Productivity and Welfare

- **Aggregate productivity**,

$$AP = \left[\int_1^{\infty} \tilde{z}^{\sigma-1} \mu(z) dz \right]^{\frac{1}{\sigma-1}},$$

where μ is the productivity distribution among active firms, and $\tilde{z}$ is realized productivity

- **Change in consumption welfare**

$$\frac{W_1^C}{W_0^C} - 1 = \left(\frac{P_0}{P_1} \right)^{\alpha} - 1,$$

where P_t is the price index for differentiated goods, and α is the expenditure share on differentiated goods

- **Change in welfare including private benefits**

$$\frac{\mathcal{W}_1}{\mathcal{W}_0} - 1 = \frac{\mathcal{I} + \mathcal{B} M_{\text{fam},1}}{\mathcal{I} + \mathcal{B} M_{\text{fam},0}} \left(\frac{P_0}{P_1} \right)^{\alpha} - 1,$$

where $\mathcal{I}_t$ is aggregate disposable monetary income, $\mathcal{B}$ is the per-firm nonpecuniary benefit of family management and $M_{\text{fam},t}$ is the mass of active family firms in period t

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